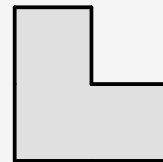


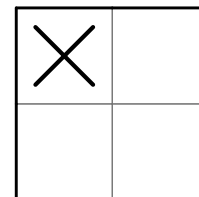
Tilings

Week 1 · Grades 4–5

A **tiling** covers a shape completely: no gaps, no overlaps, nothing sticking out past the edge. Today's piece is the **L-tromino**: three squares in an L. You may turn it and flip it any way you like.



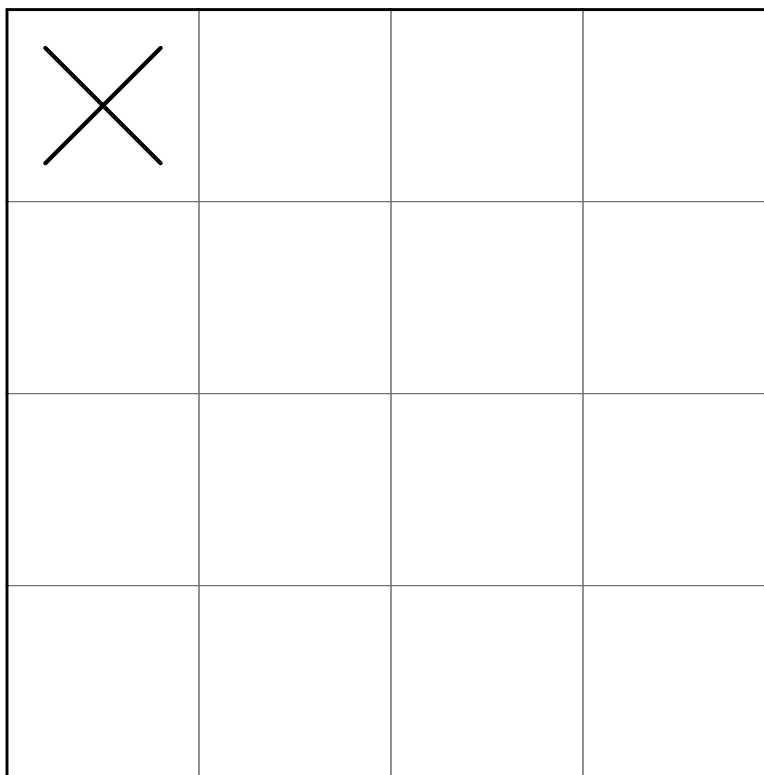
1. Warm-up



A 2×2 board with one square removed (the $\times$). How many trominoes does it take to cover the rest? _____

2. A 4×4 board with a hole

Cover every square except the $\times$ with paper trominoes. (This board is at 1-inch scale.)

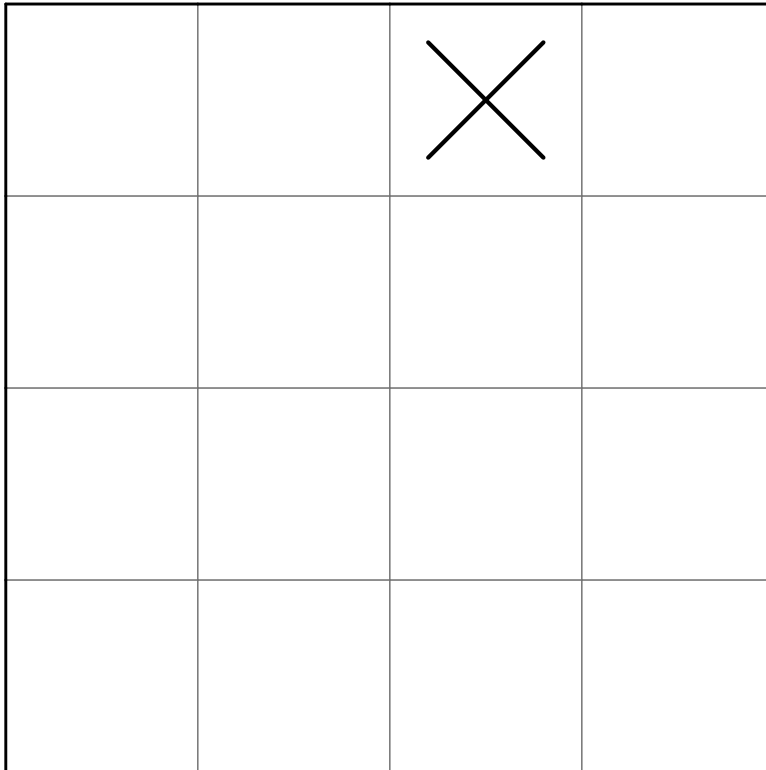


Board A

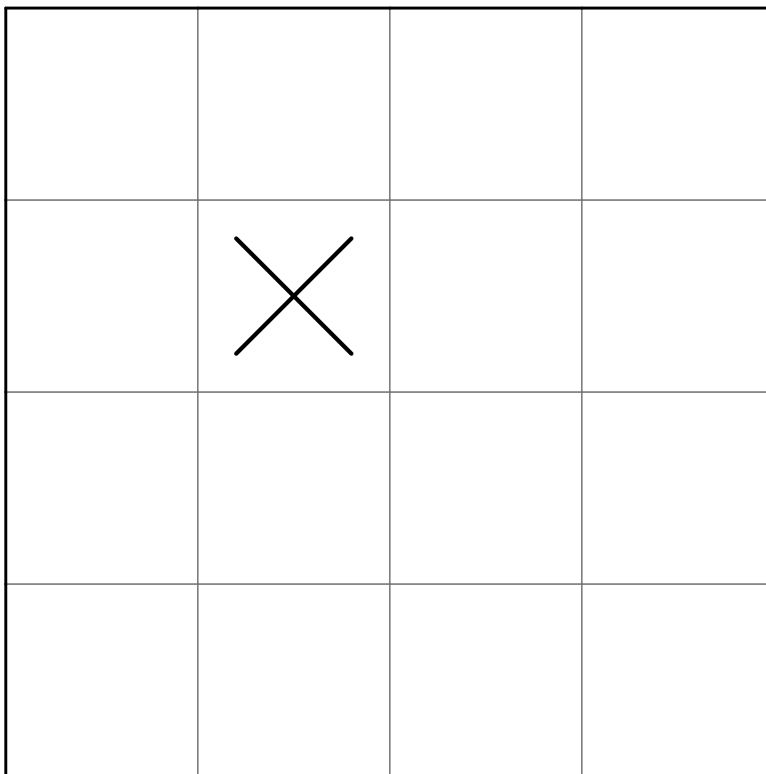
How many trominoes did you use? _____ Why could it not have been any other number?

3. Move the hole

Now the hole is on an edge (Board B) and then inside the board (Board C). Cover them.



Board B



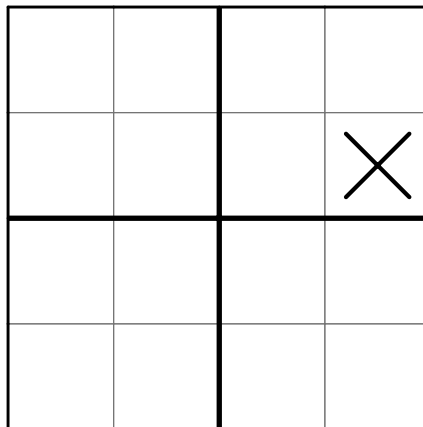
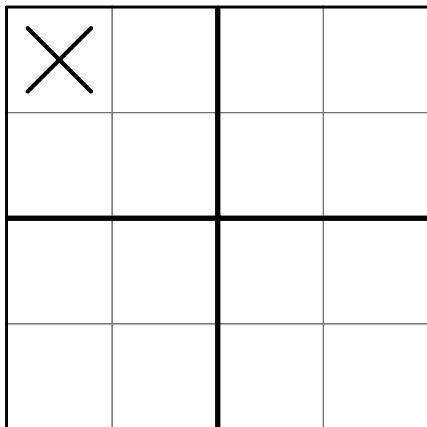
Board C

Do you think the hole can be *anywhere*? Test it: put a coin on the $\times$ of a board (so that square is back), put a second coin on some other square as the new hole, and tile around it. Try several.

Sure it always works? _____ Not sure? Turn the page.

4. A trick that always works

Two heavy lines cut the board into four 2×2 boards. The hole is in one of them. Where can you put *one* tromino so that each of the other three 2×2 boards loses exactly one square? Draw it. Then explain why that one tromino finishes the job.



5. Counting check

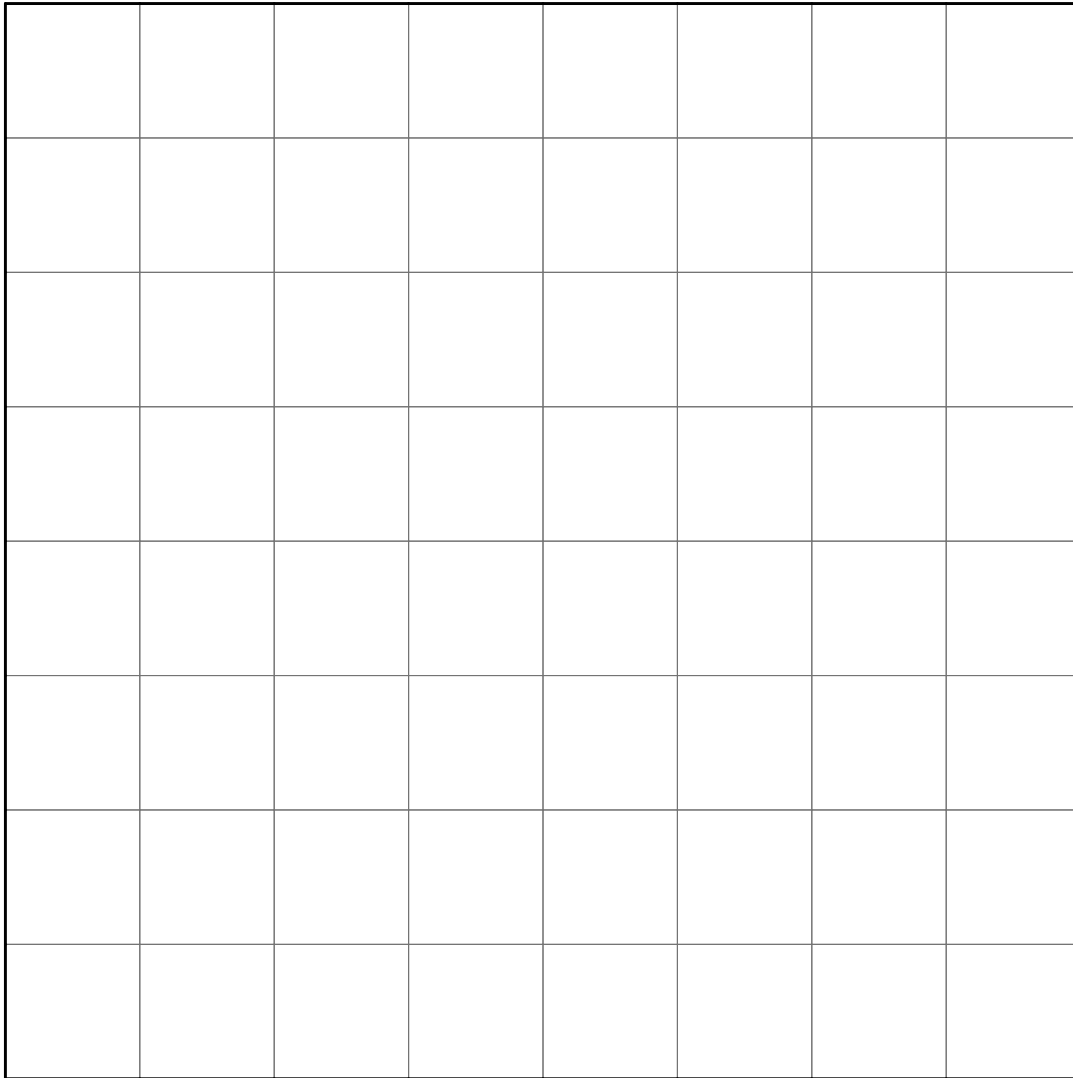
Fill in the table. Every board has one hole.

board	squares to cover	trominoes needed
2×2	3	1
4×4		
8×8		
16×16		
32×32		

What do you notice about the numbers in the middle column?

6. The 8×8 board

Mark any square you like as the hole. Then cover the rest by *drawing* the trominoes. Plan: first use the trick with four 4×4 boards. Then use the trick again inside each 4×4 board.

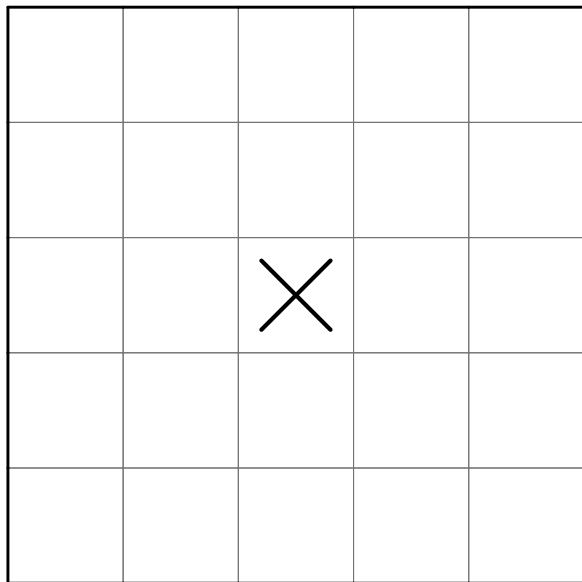


7. And then?

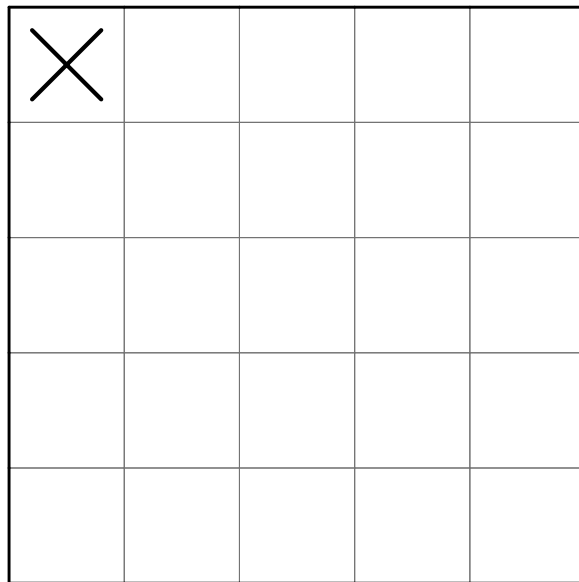
Explain to a friend why a 16×16 board with any one hole can be covered, without drawing all 85 trominoes. What about 32×32 ? Where does it stop?

8. The 5×5 board

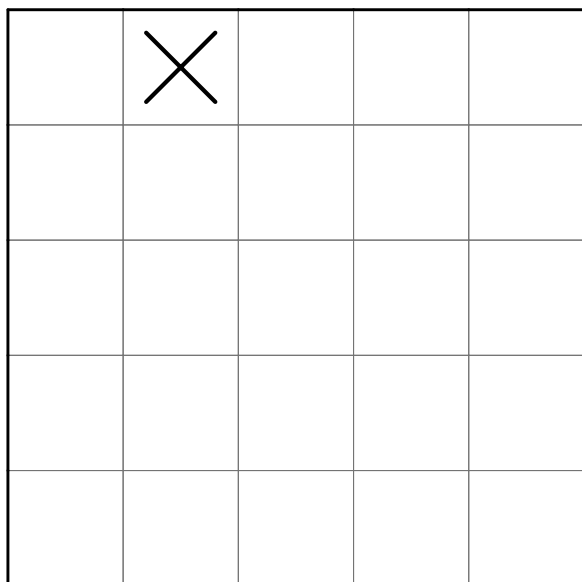
A 5×5 board with one hole has 24 squares, and $24 = 8 \times 3$, so counting does not stop us. Cover Board E (hole in the center), Board F (hole in a corner), and Board G (hole next to a corner).



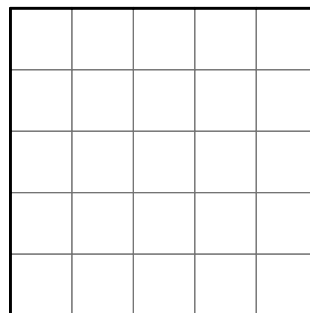
Board E



Board F



Board G

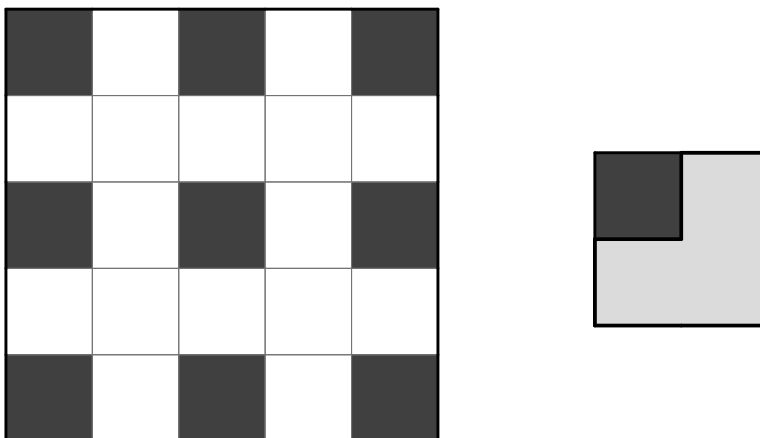


summary grid: mark • where the hole *can* be and × where it cannot

Test at least six positions of the hole before you turn the page. Draw on these boards, or mark a 5×5 board on the 1-inch grid paper, use a coin as the hole, and build with the paper trominoes.

9. Why some holes fail

Color the nine squares that are in an odd row *and* an odd column (rows 1, 3, 5 and columns 1, 3, 5) black, as shown.

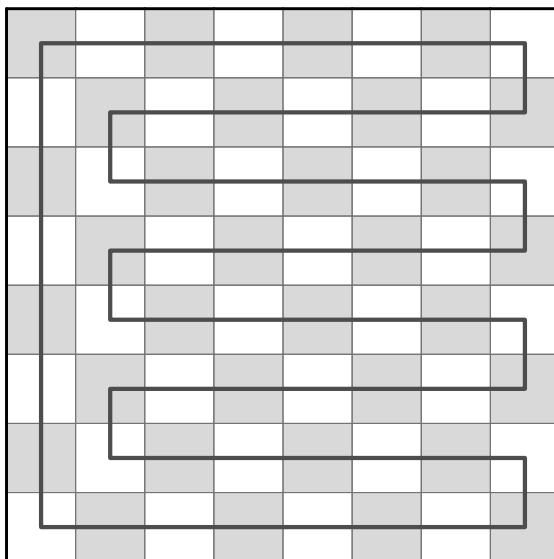


- Every tromino sits inside some 2×2 box. How many black squares can a 2×2 box contain?

- So one tromino covers at most _____ black square(s).
- Eight trominoes cover at most _____ black squares.
- If the hole is *not* a black square, how many black squares must be covered? _____
- Conclusion: _____

10. Challenge: the snake

The snake passes through every square of the chessboard once and comes back to where it started. Cross out *any* one gray square and *any* one white square. Explain why the rest can always be covered by dominoes. (What happens to the snake?)



Paper trominoes

Cut along the heavy lines. Each 3×2 block gives two L-trominoes.

